BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Multiple Choice Questions)

Note: Q.1 Choose the correct answer for each from the given options:

Q1. Mathematical expression for potentiometer is given by _____.

(a) E1/E2 = L1/L2
(b) E1/E2 = L1/L1
(c) E1 L1 = E2 L2
(d) E1 E2 = L1 L2

Q2. P.O Box is used for the measure _____.

(a) Current
(b) Voltage
(c) emf
(d) Resistance

Q3. The process in which alternative current into direct current is _____.

(a) Amplification
(b) Rectification
(c) Both a and b
(d) None of these

Q4. Momentum of photon, from the theory of relativity is related by _____.

(a) P = hc/f
(b) P = hf/c
(c) P = hcf
(d) None of these

Q5. If energy of incident photon is less then _____ no pair production will take place.

(a) 0.311 Mev
(b) 3.5 Mev
(c) $m_o c^2$
(d) $2\ m_o c^2$

Q6. The wavelength of last spectral line is Lyman series is equal to _____.

(a) R_H
(b) R^2_H
(c) 1/R_H
(d) 1/R_H

Q7. The energy of X − ray photon is _____ the energy of photon of ultra − violate light.

(a) Equal to
(b) Less than
(c) Greater than
(d) Depends on velocity of

Q8. Due to the alpha decay the mass number of atom _____.

(a) Decreased by two
(b) Decreased by four
(c) Increased by two
(d) Increased by four

Q9. Lenz's law is in accordance with the law of conservation of _____.

(a) Momentum
(b) Energy
(c) Angular Momentum
(d) All of these

Q10. The transformer in which secondary voltage is less than primary voltage is called _____.

(a) Step up Transformer
(b) Step down Transformer
(c) Rectifier
(d) Amplifier

Q11. The transformer has 1000 turns in the primary and 200 in the secondary , if input current is 2A. What will be its output current?

(a) 10 A
(b) 0.4 A
(c) 12 A
(d) None of these

Q12. _____ is a device that raises the strength of a weak signal.

(a) Transistor
(b) Amplifier
(c) Ammeter
(d) Voltmeter

Q13. Photocell is a device which converts _____ energy into _____ energy.

(a) Chemical , Light
(b) Light , Electrical
(c) Chemical , Electrical
(d) Electrical , Light

Q14. The speed of x − ray photon is always _____.

(a) Equal to C
(b) Greater than C
(c) Less than C
(d) Depends upon intensity

Q15. The Balimer series lies in the _____.

(a) Infrared region
(b) For infrared region
(c) Ultraviolet region
(d) Visible region

Q16. In any reaction $_{90}Th^{227}$ makes alpha decay the daughter element will be _____.

(a) $_{36}Kr^{39}$
(b) $_{56}Br^{144}$
(c) $_{92}U^{235}$
(d) $_{88}Ra^{223}$

Q17. The quantity of matter such that it's mass in grams in numerically equal to the _____.

(a) Molecular Formula
(b) Molecular Weight
(c) One Mole
(d) Suspected Mass

Q18. In an isobaric process the work done is equal to _____.

(a) RΔV/n
(b) C_p/nΔv
(c) C_v/nΔv
(d) nRΔT

Q19. The S.I unit of electric field intensity is _____.

(a) Newton/ Coulomb
(b) Volt/ Mater
(c) Both (a) and (b)
(d) None of these

Q20. Electron volt is the unit of _____.

(a) Power
(b) Energy
(c) Potential Difference
(d) Both a and c

Q21. What is a current passing in 5 x 10^6 ohm resistor if it is provided p.d of 5KV?

(a) 10^{-1} A
(b) 10^{-4} A
(c) 1 m A
(d) None of these

Q22. If one ampere current flow through a conductor for one hour, how much charges flow through area of cross section of conductor?

(a) 3600 C
(b) 7.2 x 10^3
(c) 1 C
(d) None of these

Q23. Tesla is the unit of _____.

(a) Magnetic Flux
(b) Electric Flux
(c) Magnetic flux Density
(d) None of these

Q24. The direction of induced current in a certain coil can easily be found by using _____.

(a) Right hand rule
(b) Faraday's law
(c) Lenz's law
(d) Both (a) and (c)

Q25. The penetrating power of neutron is _____.

(a) Zero
(b) Very low
(c) Medium
(d) Very high

Q26. The specific heat of water is _____.

(a) 4.2 x 10^3 Joule
(b) One kilo Calorie
(c) One Calore
(d) None of these

Q27. Heat can not be measured in _____.

(a) Calorie
(b) B.T.U
(c) Joule
(d) Centigrade

Q28. The Coulomb's force _____ when any material medium is palced between two charges.

(a) Increases
(b) Decreases
(c) Remains Same
(d) None of these

Q29. The Ssurface density of charge is define as charge per unit _____.

(a) Volume
(b) Length
(c) Area
(d) Mass

Q30. 1 Coloumb per one volt is equal to _____.

(a) One Joule
(b) One Ampere
(c) One Calorie
(d) One Farad

Q31. The graph between current and voltage for ohm's law is _____.

(a) Hyperbola
(b) Ellipse
(c) Straight Line
(d) Parabola

Q32. Hyb of the following is not unit of magnetic field induction is _____.

(a) Tesla
(b) Newton/Ampere Meter
(c) Weber/Meter^2
(d) Newton per volt

SECTION – B (SHORT ANSWER)

> Note: Answer any SEVEN of the following questions. Each question carries 05 marks.

Q2. Prove that co − efficient of linear expansion is one third to co − efficient of volume expansion.

Q3. Calculate electric intensity due to a sheet of infinite charges distributed uniformly.

Q4. Can you construct two wires of same length, one of copper other of iron that would have the same resistance at same temperature? Explain.

Q5. Why work done by a charge due to magnetic force is zero?

Q6. How a single device is used to measure the resistance , current and potential?

Q7. Why are the energy levels of hydrogen atom less than zero?

Q8. How the photon is differe from the matter?

Q9. The battery of a pocket calculator supplies 0.25 A at a p.d of 8V. What is the power rating of the calculator?

Q10. Calculate the energy of x − ray photon having wavelength 2 x 10^{-10}m.

Q11. What is the wavelength of the radiation that is emitted when a hydrogen atom undergoes a transition from the state n = 3 to n = 1.

SECTION – C (DESCRIPTIVE ANSWER)

> Note: Answer any TWO of the following questions. Each question carries 09 (5 + 4) marks.

Q12. (a)What is specific heat and molar specific heat? Show that Cp−Cv=R. (b) Describe Bohr's atomic Model. Hence calculate the energy in n^{th} orbit.

Q13. (a) What is charge to mass ratio of an electron? How it calculated by different two methods? (b) What are the assumptions of theory of relativity? And write the results of special theory of relativity.

Q14. Write the notes on any TWO of the following. (i) Wheat Sstone Bbridge (ii) Power Ddissipation in Rresistor (iii) Nuclear Fission and Fusion (iv) A.C Generator